UDRF-SI APPLICATION TEMPLATE AND INSTRUCTIONS

Please complete the following form when applying to the University of Delaware Research Foundation Strategic Initiatives (UDRF-SI) seed grant program. All UDRF-SI proposals should be single-spaced using 11-point Arial font with one-inch margins.

PLEASE NOTE: Applicants do not need to generate a PeopleSoft proposal at the time of proposal submission. If the proposal is selected for funding, detailed instructions for PeopleSoft submission will follow.

Questions may be directed to **researchdev@udel.edu.**

SUBMISSION CHECKLIST:

- ☑ Completion of this template, including preliminary budget information
- ☑ Executive Summary (Maximum one-page)
- ☑ Project Description (Maximum three-pages)
- ☑ References (Maximum two-pages)
- ☑ Budget Justification (Maximum one-page)
- ☑ Appendix for PI's and Co-PI's biosketches (NIH or NSF-format, Maximum 3 pages ea.)
- ☑ Appendix for PI's and Co-PI's Current & Pending Support documents

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Name, department/college of the principal investigator:

Dr. Jennifer Bigman
School of Marine Science and Policy
College of Earth, Ocean, and Environment

Please list the name(s) of any Research Institute(s) involved in the project.

NA

Name(s), department(s)/college(s) of UD co-investigators (Co-PIs must be at UD):

Dr. Matthew Oliver, School of Marine Science and Policy, College of Earth, Ocean, and Environment
Dr. Aaron Carlisle, School of Marine Science and Policy, College of Earth, Ocean, and

Title of the project:

Building a dynamic ocean management tool to minimize fisheries bycatch along the U.S. East Coast

Detail any new space, facilities, and/or equipment needs (incl. renovations) associated with the project:

NA

BUDGET

Complete this budget table. A detailed yearly budget will be requested if the proposal is awarded. For budget considerations and restrictions, refer to the published UDRF-SI guidelines on the Research Office website. Maximum budget allowed is $55,000.

Budget Category	Cost
Graduate Student Stipend/Wage	50,419
Undergrad Student Wage	
Other Personnel Salary [PI & Co-PI(s) unallowed]	
Fringe Benefits	3,731
Supplies, Equipment and Materials	
Travel	850
Participant Support Costs	
Other Direct Costs (specify)	
Total Direct Project Costs	55,000
PI Faculty Cost-share Effort (%, [5% FTE minimum])	8.33%

PROPOSAL COMPONENTS

For detailed information about the Executive Summary and Project Description requirements, refer to the published **UDRF-SI guidelines** found on the Research Office website. Additional information such as the References, Budget Justification, biosketches, and Current & Pending Support documents should be pasted in or combined with this proposal template. The proposal package should be submitted as one PDF file. Incomplete applications will be returned without review.

Required Sections:

- Executive Summary (Maximum one-page; must include Overview, Research Significance and Approach, and Broader Impacts & Future Funding)
- Project Description (Maximum three-pages; must include a timeline and a brief paragraph describing the Mentoring Plan for mentoring the untenured PI)
- References (Maximum two-pages): Use the format accepted for bibliographic references in your discipline
- Budget Justification (Maximum one-page): Provide detailed justification for budget items listed in the budget table. Find **UD's budget justification template** here.
- Appendix, PI and Co-PI Biosketches (NIH or NSF-format maximum three-pages each)
- Appendix, PI and Co-PI Current & Pending Support: Include all current and pending support, including this project, for ongoing projects, and any proposals currently under consideration (excluding departmental start-up funds). Each project entry should include the following information:
  - "Project Title"
  - Status of Support: Current, Pending, or Planned Source of Support
  - Project/Proposal Start and End Dates
  - Total Award Amount

Bigman UDRF-SI 2026
Executive Summary

Overview

Bycatch, the unintended capture of nontarget species in fisheries, depletes marine populations, alters ecosystem structure and function, increases fishing costs, and restricts fishing opportunities. Reducing these impacts is especially challenging where fisheries share fishing grounds and species targeted in one fishery are caught incidentally in another. Environmental change further complicates bycatch reduction efforts by shifting species' distributions and altering where target and nontarget species overlap. We propose to develop a novel, dynamic bycatch reduction tool for high-value U.S. East Coast fisheries while addressing fundamental ecological questions about how environmental conditions shape species' distributions across space and time. Our aims are to (1) identify environmental conditions associated with concentrations of target and bycatch species and (2) quantify how their distributions overlap and vary together. Joint species distribution models will produce predictive maps identifying opportunities to reduce bycatch while maintaining fishing opportunities for target species. This work lays the foundation for a tool that provides daily guidance on where to fish as ocean conditions change, aligning predictions with the scales of species movement, fishing behavior, and fisheries management.

Research significance and approach

Understanding factors that govern where and when species concentrate and co-occur is essential for predicting ecological responses to environmental change and sustaining fisheries, healthy ecosystems, and the livelihoods and communities they support. The proposed project will examine how distributions of target and bycatch species overlap and vary together as environmental conditions change, and how these patterns can inform management that maintains fishing opportunities across interconnected fisheries. We will focus on eight species targeted, caught incidentally, or both in New England and Mid-Atlantic trawl fisheries, combining over five decades of fish abundance and biomass observations from publicly available fisheries monitoring surveys with environmental data derived from ocean models and satellites. Novel joint species distribution models will estimate how environmental conditions relate to each species' density and whether species concentrate in the same places or change together over time. These models will quantify confidence in identifying areas with high target-species density and low bycatch-species density. We will evaluate model predictive performance using cross-validation and assess how potential bycatch avoidance opportunities vary across locations, seasons, years, and fisheries. We will also test sensitivity to thresholds defining high and low bycatch density, identifying where opportunities are consistently supported and where uncertainty is warranted.

Broader Impacts and Future Funding

The Mid-Atlantic Fishery Management Council (Council) develops and recommends management measures to sustain fisheries along the U.S East Coast, efforts that require information on where target species can be caught with minimal bycatch as ocean conditions change. The proposed project will produce predictive surfaces identifying areas with high target-species density and low bycatch-species density, laying the foundation for daily maps that can help fishers decide where to fish and help managers evaluate bycatch mitigation measures. Engagement with fishers, other scientists, and state and federal agencies will help ensure these products support practical fishing and management decisions. Jennifer's membership on the Council's Scientific and Statistical Committee, which advises the Council on the best available science, provides a direct pathway for sharing findings with decision-makers. The project will also train a graduate student in ecology and statistical modeling, and support development of a teaching module for Jennifer's UD marine ecology course. Preliminary findings and analytical framework developed herein will form the basis for external proposals that continue development and evaluation of a dynamic bycatch reduction tool (e.g., adding fishing-vessel catch data) accessible through a web or mobile application. Planned funding targets include NASA, NFWF, and NOAA, with proposals prepared and submitted during the UDRF-SI award period.

Bigman UDRF-SI 2026
Project Description

Aims and Significance

Global capture fisheries generate around US$159 billion annually, but population declines threaten these economic benefits and associated livelihoods [1], [2]. Sustaining these resources requires harvesting at rates that allow populations to replenish while preserving the ecosystems that support them [3], [4]. However, fisheries management largely regulates individual target species, with limited consideration of environmental and interspecific relationships [5], [6]. As a result, management outcomes are inconsistent: some populations remain productive or have recovered from heavy exploitation, while others remain depleted or are in decline [7], [8].

Bycatch, the incidental capture of nontarget species, imposes economic and ecological costs [9], [10]. It increases handling time, restricts fishing opportunities, and can reduce future catches, while depleting nontarget populations and altering ecosystems [11], [12], [13], [14]. Mitigation includes fixed time-area closures, bycatch limits, and gear modifications, among others, but effectiveness depends on identifying where and when bycatch poses the greatest concern, which varies among species, fisheries, gear types, and regions [15], [16], [17]. These efforts are complicated by fisheries operating in the same area, targeting multiple species, and catching species incidentally that are targeted in other fisheries [10], [18].

Environmental change compounds this challenge by shifting species' habitat use and distributions, altering where and when target and nontarget species overlap and reducing efficacy of bycatch mitigation measures [17]. These changes are particularly relevant in waters off the U.S. East Coast, where many marine species have already shifted northward in response to warming, and continued changes are projected to reduce fishing opportunities for communities historically dependent on these resources [19], [20]. **A key gap in reducing bycatch is understanding how distributions of target and nontarget species overlap and change together with shifting environmental conditions, and how these patterns can inform management that maintains fishing opportunities across interconnected fisheries.**

We propose the development of a novel, dynamic bycatch reduction tool for high-value U.S. East Coast fisheries. Integrating catch, survey, and environmental data, we will predict areas with high expected target catch and low bycatch risk. These maps will support evaluation of currently implemented bycatch mitigation measures, while laying the foundation for daily predictive maps that align species movement, fishing, and management. **Our aims are to: (1) identify environmental conditions associated with target and bycatch hotspots and (2) quantify how their distributions overlap and vary together across space and time.**

Approach and Timeline

We will begin development of a novel, dynamic bycatch reduction tool by identifying environmental conditions associated with high concentrations of target and bycatch species and quantifying how distributions overlap and change together over time. We will focus on eight species: black sea bass, summer flounder, scup, longfin squid, butterfish, spiny dogfish, striped sea robin, and northern sea robin. These species are targeted, caught incidentally, or both in New England and Mid-Atlantic large and small mesh bottom otter-trawl fisheries, which have some of the highest bycatch rates in the U.S. [10], [12], [21].

We will use publicly available long-term (> 50 years) spring and fall bottom trawl survey catch data from NOAA Fisheries to estimate relative biomass and numerical density (kg or individuals/km² swept, respectively) [22], [23]. Environmental data from ocean models and satellite products will include bottom temperature, dissolved oxygen, chlorophyll-a concentration (proxy for primary productivity), and sea surface height [24], [25]. Joint species distribution models

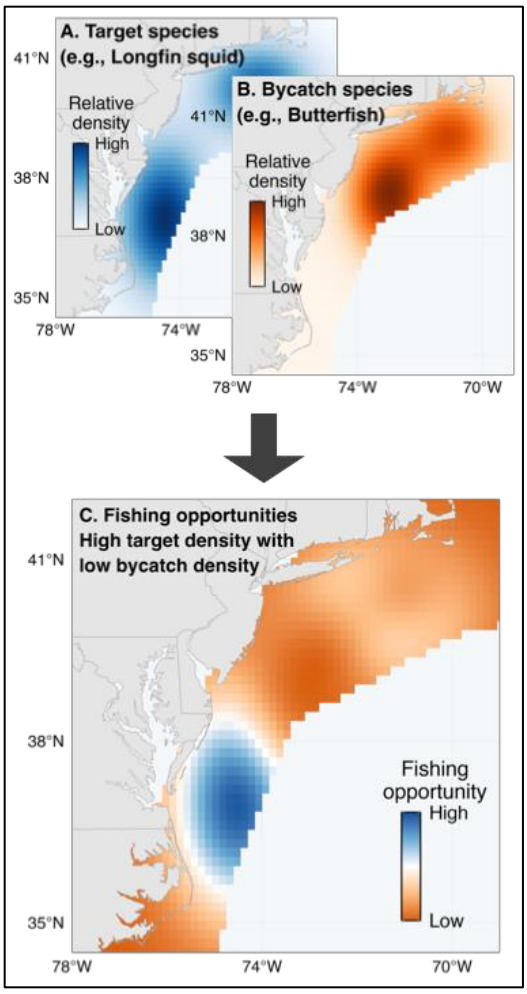


Figure 1: Conceptual illustration of proposed bycatch reduction tool. Simulated distributions of a (A) target and (B) bycatch species are combined to identify (C) fishing opportunities with high target but low bycatch density. The tool will be comprised of daily maps that align scales of species movement, fishing, and management.

will model density as a function of the environment, season, and spatial and spatiotemporal random fields, which represent persistent geographic patterns and local changes in density unexplained by other factors in the model [26], [27]. Associations among species' spatial patterns will quantify whether their densities vary together after accounting for environmental conditions and will be propagated into joint predictions and uncertainty to assess confidence in identifying areas with high target and low bycatch density [28], [29]. Cross-validation will assess model convergence, parameter uncertainty, residual spatial/temporal structure, and predictive performance [28], [29]. Species-specific and joint-species maps will characterize co-occurrence and bycatch-avoidance opportunities for a given combination (Figure 1). We will evaluate how opportunities vary across space and time and sensitivity to thresholds defining high/low densities.

All code and data will be publicly available on GitHub and archived with persistent digital identifiers. Jennifer is active in the open science movement and teaches open science and reproducible research skills (e.g., coding, git, GitHub) at UD.

Timeline:

Activity	Year 1				Year 2			
Project planning including first team meeting	■							
Access and clean fishery and environmental datasets	■	■	■					
Statistical model development, inspection, and validation			■	■				
Paper/thesis writing					■	■	■	■
Engage with state/federal agencies, MAC-AFS conference	■				■			■
Apply for external awards based on this project	■		■		■		■	
Develop teaching module for UD courses							■	■
Regular meeting with mentors and student	■	■	■	■	■	■	■	■

Career Development and Mentoring

Dr. Jennifer Bigman is a quantitative ecologist who integrates theory, statistics, and diverse data streams to understand ecological patterns. The work proposed here builds on her expertise in developing quantitative methods and assessing biological responses to environmental variability while expanding her research trajectory into dynamic ocean management, drawing on her co-PIs' expertise. The project will establish the methodological foundation and preliminary results for a broader research program linking theory and applied

science to quantify and forecast distribution shifts and their effects on ecosystems and the fishing industries and communities they support. Predictions of species' concentrations and overlap serve as a foundation that positions Jennifer to pursue external funding to continue development, evaluation, and implementation of a novel, dynamic bycatch reduction tool.

Dr. Matthew Oliver is an expert in ocean observing, remote sensing, and biological oceanography, and has active projects focused on building a tool to predict changes in Atlantic sturgeon habitat use from satellite-derived environmental data. He will provide mentorship in these topics, as well as proposal writing and securing external funding. Dr. Aaron Carlisle is an expert in spatial ecology, fish physiology, and East Coast fish and fisheries, and will provide mentorship in these topics, as well as connect Jennifer with DE state agencies to network and seek additional funding opportunities. Matthew and Aaron have secured external funding from state and federal agencies (NASA, NOAA, BOEM, NSF) and nonprofits and have mentored early career faculty, having served as official UD faculty mentors, as well as postdoctoral supervisors.

The team will meet regularly to discuss plans, progress, technical details, environmental data usage, spatial ecology, among other topics (once a month). Matthew and Aaron will serve as committee members of the graduate student and have regular meetings to ensure student progress and development (once a quarter). Project milestones include (1) compiling and cleaning datasets, (2) developing joint species distribution modeling framework, (3) generating predictive maps and comparing with existing bycatch mitigation measures, (4) completing and submitting a manuscript to a scientific journal, and (5) write proposals for external funding.

Broader Impacts and Funding Trajectory

This project will provide fishers and natural resource managers much-needed information to reduce bycatch while maintaining fishing opportunities in Mid-Atlantic and New England bottom trawl fisheries, promoting the sustainable use of living marine resources. Predictive maps will lay the foundation to deliver daily guidance on fishing opportunities through a publicly accessible web or mobile application. Other scientists, fishers, and state/federal agencies will be engaged to ensure outputs address management needs. For example, this work addresses specific management needs of the Mid-Atlantic Fishery Management Council, which is mandated by law to develop fishery management plans, or frameworks for conserving and managing fisheries, and recommends catch limits and harvest measures for federally managed species along the U.S. East Coast including butterfish, longfin squid, black sea bass, summer flounder, spiny dogfish, and scup [30]. Jennifer serves on the Council's Scientific and Statistical Committee, which provides scientific advice to support management decisions, creating a direct pathway for communicating project findings and informing fisheries management.

The preliminary results, modeling framework, and predictive maps from this project will support proposals prepared and submitted during the UDRF-SI award period. The next phase will integrate fishing-vessel catch data to distinguish species co-occurrence from realized fisheries catch and evaluate how these relationships vary among locations, species, gear types, and fisheries. Planned funding targets are NASA's A.15 Biosphere solicitation ("Predicting fishing opportunities from space", 3 yr., ~ $900,000); NFWF's Delaware Watershed Conservation Fund ("Predicting fisheries bycatch in and around Delaware Bay", 3 yr., ~ $500,000), and NOAA's National Bycatch Reduction Engineering Program ("A novel bycatch reduction tool for sustainable fisheries management along the U.S. East Coast", 18 mo., ~ $400,000).

Additional outcomes include training of a graduate student and developing a teaching module for Marine Ecology (MAST 316), a required undergraduate course that Jennifer teaches each spring at UD. The module will build on a semester-long project in which students compile occurrence records from global databases and map changes in species distributions. Drawing on findings from this UDRF-SI project, students will analyze environmental conditions associated with distributional changes and explore their societal and ecological consequences, including effects on fishing communities and bycatch risk.

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[4] J. S. Link, “What Does Ecosystem-Based Fisheries Management Mean?,” vol. 27, no. 4, 2002.

[5] M. Skern-Mauritzen *et al.*, “Ecosystem processes are rarely included in tactical fisheries management,” *Fish Fish.*, vol. 17, no. 1, pp. 165–175, Mar. 2016, doi: 10.1111/faf.12111.

[6] K. N. Marshall, L. E. Koehn, P. S. Levin, T. E. Essington, and O. P. Jensen, “Inclusion of ecosystem information in US fish stock assessments suggests progress toward ecosystem-based fisheries management,” *ICES J. Mar. Sci.*, vol. 76, no. 1, pp. 1–9, Jan. 2019, doi: 10.1093/icesjms/fsy152.

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Bigman UDRF-SI 2026
Budget Justification

Salaries and Wages ($50,419 sponsor/match, $24,026 cost share)

Senior Personnel ($0 sponsor/match, $24,026 cost share)
Dr. Jennifer Bigman will serve as the PI of the project and will oversee the entire project, including startup/planning, technical, administrative, project reporting, etc. She will provide 1.0 month of effort (8.33% FTE) per year in the form of department cost share. She will hire and advise the graduate student working on species distribution modeling of target and bycatch fish species. Dr. Bigman will also organize and lead all team meetings, ensure the team and activities/outcomes are on track, ensure project deliverables are completed, and lead engagement with state and federal agencies including for data access, collaboration, and communicating project outcomes. She will guide project communication, data analysis, manuscript preparation and dissemination of results, activities that will be led by the graduate student. Salaries are increased by 4% annually on September 1st.

Graduate Student ($50,419 sponsor/match, $0 cost share)
Funds are requested for 12 months of support in year 1 and 4 months of support in year 2 for a graduate student to lead project communication, data analysis, manuscript preparation and dissemination of results, guided by the project PI. The annual salary amount for a Graduate Student in the College of Earth, Ocean and Environment, School of Marine Science & Policy, is $36,000 for 12 months of effort effective 9/1/2026. We anticipate a 4% annual increase thereafter each September 1st.

Fringe ($3,731 sponsor/match, $10,547 cost share)
According to University policy, fringe benefits are calculated as 43.9% for faculty and professional Employees, 7.4% for graduate students, 71.0% for Salaried Staff, and 5.5% for undergraduate students and miscellaneous wage employees.

Other Direct Costs ($850 sponsor/match, $0 cost share)
Travel – $850 is requested in year 2 to help support conference travel expenses for the PI or graduate student working on the project. The PI or graduate student will present this work at the annual meeting of the American Fisheries Society, which is hosted domestically and alternates between the eastern and western U.S. each year. Travel costs may include, but are not limited to, airfare, baggage fees, ground transportation fees, parking fees, conference registration fees, lodging, meals and incidentals. Meals and incidentals will be reimbursed at the GSA rate for the destination city. Any mileage reimbursement will be at the GSA rate, currently $0.76/mile.

Matching ($20,000)
Matching funds of $20,000 will be provided by the Provost's office ($10,000), the College of Earth, Ocean, and Environment (CEOE) ($5,000) and CEOE's School of Marine Science and Policy ($5,000) per the requirements of the solicitation and are included in the budget.

The University of Delaware will consider the costs and requests in the justification approved if an award is made and no contrary guidance from the agency is included in the award notice.

IDENTIFYING INFORMATION:

NAME: Bigman, Jennifer

ORCID iD: https://orcid.org/0000-0001-8070-3061

POSITION TITLE: Assistant Professor

PRIMARY ORGANIZATION AND LOCATION: University of Delaware, School of Marine Science and Policy, Lewes, Delaware, United States

Professional Preparation:

ORGANIZATION AND LOCATION	DEGREE (if applicable)	RECEIPT DATE	FIELD OF STUDY
National Science Foundation, NOAA Alaska Fisheries Science Center, Seattle, Washington, United States	Postdoctoral Fellow	06/2022 - 05/2024	Marine Biology, Fisheries Science
NOAA Alaska Fisheries Science Center, Seattle, Washington, United States	Postdoctoral Fellow	07/2021 - 05/2022	Marine Biology, Fisheries Science
Simon Fraser University, Burnaby, British Columbia, Canada	PHD	01/2015 - 06/2021	Marine Biology, Biology
Moss Landing Marine Labs, Moss Landing, California, United States	MS	08/2009 - 07/2013	Marine Biology, Biology
University of North Carolina Wilmington, Wilmington, North Carolina, United States	BS	08/2005 - 05/2009	Marine Science, Marine Biology

Appointments and Positions

2025 - present Assistant Professor, University of Delaware, School of Marine Science and Policy, Lewes, Delaware, United States

2026 - present Member, Mid-Atlantic Fishery Management Council Scientific and Statistical Committee, Dover, Delaware, United States

2025 - present Co-Chair, Working Group on Impacts of Warming on Growth Rates and Fisheries Yields, International Council for Exploration of the Seas, N/A, Not Applicable, N/A, N/A

2024 - 2025 Fish Biologist, NOAA Office of Science and Technology, Seattle, Washington, United States

2020 - present Certified Instructor, Software Carpentries, N/A, Not Applicable, N/A, N/A

2014 - 2014 Fellow, Delta Science Program, California Sea Grant, Sacramento, California, United States

2013 - 2021	Northeast Pacific Regional Working Group Member, Shark Specialist Group, International Union for the Conservation of Nature, N/A, Not Applicable, N/A, N/A
2013 - 2014	Research Technician, Alaskan Skate Trophic Ecology, University of Washington, Seattle, Washington, United States
2012 - 2012	Research Technician, Monterey Bay Isoscape, Moss Landing Marine Labs, Moss Landing, California, United States
2011 - 2012	Research Technician, Geological Oceanography Lab, Moss Landing Marine Labs, Moss Landing, California, United States
2011 - 2012	Research Technician, Basking Shark Sightings Network, Moss Landing Marine Labs, Moss Landing, California, United States
2008 - 2008	Intern, Sea Turtle Conservation and Research Program, Mote Marine Laboratory, Sarasota, Florida, United States
2006 - 2009	Research Technician, Benthic Ecology Laboratory, University of North Carolina Wilmington, Wilmington, North Carolina, United States

Products

Products Most Closely Related to the Proposed Project

1. Jennifer S. Bigman, Lewis A.K. Barnett, James T. Thorson, Sean C. Anderson, Krista Oke, Kelly Kearney, Darren Pilcher, Wei Cheng, Esther Goldstein, Mary E. Matta, Kirstin K. Holsman, Lauren A. Rogers. Changes in body size with age do not follow the temperature-size rule. 2025 September. DOI: 10.1101/2025.09.02.673738
2. Audzijonyte A, Andersen K, Atkinson D, Bigman J, Blanchard J, Coghlan A, Heather F, Lindmark M, Morrongiello J, Pauly D. Which Body Size Metrics Should Be Used for Assessing Temperature Impacts on Fish Growth and Size?. Global Change Biology. 2025 June 18; 31(6):-. Available from: https://onlinelibrary.wiley.com/doi/10.1111/gcb.70296 DOI: 10.1111/gcb.70296
3. Bigman J, Laurel B, Kearney K, Hermann A, Cheng W, Holsman K, Rogers L. Predicting Pacific cod thermal spawning habitat in a changing climate. ICES Journal of Marine Science. 2025 January; 82(1):-. Available from: https://academic.oup.com/icesjms/article/doi/10.1093/icesjms/fsad096/7204287 DOI: 10.1093/icesjms/fsad096
4. Bigman J, Kearney K, Holsman K. Bias correction for integrated climate projection modeling and its effect on downstream biological models. [Preprint]. 2026 January 14. DOI: 10.64898/2026.01.12.698884
5. Rogers LA, Axler KE, Bigman JS. Widespread phenological shifts with temperature in Alaska's marine fishes. Global Change Biology. 2026 January 16; 32. Available from: https://onlinelibrary.wiley.com/doi/epdf/10.1111/gcb.70708 DOI: https://doi.org/10.1111/gcb.70708

Other Significant Products, Whether or Not Related to the Proposed Project

1. Thorson J, Maureaud A, Frelat R, Mérigot B, Bigman J, Friedman S, Palomares M, Pinsky M, Price S, Wainwright P. Identifying direct and indirect associations among traits by merging

phylogenetic comparative methods and structural equation models. Methods in Ecology and Evolution. 2023 March 26; 14(5):1259-1275. Available from: https://besjournals.onlinelibrary.wiley.com/doi/10.1111/2041-210X.14076 DOI: 10.1111/2041-210X.14076

2. Bigman J, M'Gonigle L, Wegner N, Dulvy N. Respiratory capacity is twice as important as temperature in explaining patterns of metabolic rate across the vertebrate tree of life. Science Advances. 2021 May 07; 7(19):-. Available from: https://www.science.org/doi/10.1126/sciadv.abe5163 DOI: 10.1126/sciadv.abe5163
3. Barrowclift E, Bigman J, Digel E, Berggren P, Dulvy N. Offspring Size Resolves a Population Growth Paradox in Rays and Skates. Fish and Fisheries. 2026 May 12; 27(4):927-941. Available from: https://onlinelibrary.wiley.com/doi/10.1111/faf.70091 DOI: 10.1111/faf.70091
4. Bigman JS, Wegner NC, Dulvy NK. Revisiting a central prediction of the Gill Oxygen Limitation Theory: Gill area index and growth performance. Fish and Fisheries. 2023 February 27; 2023(00):1-13. Available from: https://onlinelibrary.wiley.com/doi/epdf/10.1111/faf.12730 DOI: 10.1111/faf.12730
5. Gravel S, Bigman J, Pardo S, Wong S, Dulvy N. Metabolism, population growth, and the fast-slow life history continuum of marine fishes. Fish and Fisheries. 2024 January 11; 25(2):349-361. Available from: https://onlinelibrary.wiley.com/doi/10.1111/faf.12811 DOI: 10.1111/faf.12811

Certification:

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Certified by Bigman, Jennifer in SciENcv on 2026-10-22 16:29:21

IDENTIFYING INFORMATION:

NAME: Oliver, Matthew

ORCID iD: https://orcid.org/0000-0002-6576-914X

POSITION TITLE: Patricia & Charles Robertson Distinguished Professor of Marine Science & Policy

PRIMARY ORGANIZATION AND LOCATION: University of Delaware, Lewes, Delaware, United States

Professional Preparation:

ORGANIZATION AND LOCATION	DEGREE (if applicable)	RECEIPT DATE	FIELD OF STUDY
Rutgers University, New Brunswick, New Jersey, United States	PHD	09/2001 - 10/2006	Oceanography
California Polytechnic State University, San Luis Obsipo, California, United States	MS	09/1999 - 06/2001	Biology
California Polytechnic State University, San Luis Obispo, California, United States	BS	01/1996 - 06/1999	Ecology and Systematic Biology
Cerritos College, Norwalk, California, United States	AA	09/1993 - 12/1995	Natural Sciences

Appointments and Positions

2020 - present Patricia & Charles Robertson Distinguished Professor of Marine Science & Policy, University of Delaware, Lewes, Delaware, United States

2015 - 2020 Patricia & Charles Robertson Associate Professor of Marine Science & Policy, University of Delaware, Lewes, Delaware, United States

2014 - 2015 Associate Professor of Oceanography, University of Delaware, Lewes, Delaware, United States

2008 - 2014 Assistant Professor of Oceanography, University of Delaware, Lewes, Delaware, United States

2007 - 2007 Postdoctoral Researcher, Rutgers University, New Brunswick, New Jersey, United States

Products

Products Most Closely Related to the Proposed Project

1. Pinti J, Carlisle AB, Huntley HS, Shatley M, Block BA, Oliver MJ. The role of oceanographic scales in shaping highly mobile marine predator distributions. Sci Rep. 2025 Jul 2;15(1):23235. PubMed Central PMCID: PMC12222499.

2. Pinti J, Shatley M, Huntley H, Carlisle A, Oliver M. The interplay between animal location accuracy and the decorrelation length scale of environmental variables when investigating environmental selection in marine organisms. Marine Ecology Progress Series. 2024; 732:1-14. issn: 0171-8630
3. Pinti J, Shatley M, Carlisle A, Block BA, Oliver MJ. Using pseudo-absence models to test for environmental selection in marine movement ecology: the importance of sample size and selection strength. Mov Ecol. 2022 Dec 29;10(1):60. PubMed Central PMCID: PMC9798696.
4. Hobday A, Hartog J, Manderson J, Mills K, Oliver M, Pershing A, Siedlecki S. Ethical considerations and unanticipated consequences associated with ecological forecasting for marine resources. ICES Journal of Marine Science. 2019; 76(5):1244-1256. issn: 1054-3139
5. Breece M, Fox D, Haulsee D, Wirgin I, Oliver M. Satellite driven distribution models of endangered Atlantic sturgeon occurrence in the mid-Atlantic Bight. ICES Journal of Marine Science. 2018; 75(2):562-571. issn: 1054-3139

Other Significant Products, Whether or Not Related to the Proposed Project

1. Ogburn M, Bowers M, Stephenson G, Curtis T, Davis M, Fox D, Hale E, Horne A, Iafrate J, Mathies N, Oliver M, Richie K, Secor D, Skomal G, Stence C, Winton M. Applications of acoustic telemetry to management and conservation in the Atlantic Cooperative Telemetry Network. Can. J. Fish. Aquat. Sci.. 2026 January 01; 83:1-15. Available from: https://doi.org/10.1139/cjfas-2025-0349 DOI: 10.1139/cjfas-2025-0349
2. Breece MW, Oliver MJ, Fox DA, Hale EA, Haulsee DE, Shatley M, Bograd SJ, Hazen EL, Welch H. A satellite-based mobile warning system to reduce interactions with an endangered species. Ecol Appl. 2021 Sep;31(6):e02358. PubMed Central PMCID: PMC8459280.
3. Welch H, Brodie S, Jacox M, Robinson D, Wilson C, Bograd S, Oliver M, Hazen E. Considerations for transferring an operational dynamic ocean management tool between ocean color products. Remote sensing of Environment. 2020; 242:111753. issn: 0034-4257
4. Rothermel ER, Balazik MT, Best JE, Breece MW, Fox DA, Gahagan BI, Haulsee DE, Higgs AL, O'Brien MHP, Oliver MJ, Park IA, Secor DH. Comparative migration ecology of striped bass and Atlantic sturgeon in the US Southern mid-Atlantic bight flyway. PLoS One. 2020;15(6):e0234442. PubMed Central PMCID: PMC7299546.
5. Haulsee D, Breece M, Brown L, Wetherbee B, Fox D, Oliver M. Spatial ecology of Carcharias taurus in the northwestern Mid-Atlantic coastal ocean. Marine Ecology Progress Series. 2018; 597:191-206. issn: 0171-8630

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Certified by Oliver, Matthew in SciENcv on 2026-10-22 17:10:56

IDENTIFYING INFORMATION:

NAME: Carlisle, Aaron

ORCID iD: https://orcid.org/0000-0003-0796-6564

POSITION TITLE: Associate Professor

PRIMARY ORGANIZATION AND LOCATION: University of Delaware, Lewes, Delaware, USA

Professional Preparation:

ORGANIZATION AND LOCATION	DEGREE (if applicable)	RECEIPT DATE	FIELD OF STUDY
Stanford University/NOAA, Pacific Grove, California, United States	Postdoctoral Fellow	07/2012 - 12/2017	Marine Ecology
Stanford University, Pacific Grove, California, United States	PHD	09/2006 - 07/2012	Biological Sciences
Moss Landing Marine Laboratories/San Jose State University, Moss Landing, California, United States	MS	08/2000 - 09/2006	Marine Science
Princeton University, Princeton, New Jersey, United States	AB	09/1995 - 05/1999	Ecology and Evolutionary Biology

Appointments and Positions

2018 - present Associate Professor, University of Delaware, Lewes, Delaware, USA
2024 - 2025 Associate Professor, University of Delaware, Lewes, Delaware, United States
2018 - 2024 Assistant Professor, University of Delaware, Lewes, Delaware, United States

Products

Products Most Closely Related to the Proposed Project

1. Arnoldi N, Carlisle A, Andrzejaczek S, Castleton M, Micheli F, Schallert R, White T, Block B. Salmon shark seasonal site fidelity demonstrates the influence of scale on identifying potential high-use areas and vulnerabilities. Marine Ecology Progress Series. 2024 May 02; 735:125-140. Available from: https://www.int-res.com/abstracts/meps/v735/p125-140/ DOI: 10.3354/meps14565
2. Jacoby D, Ferretti F, Freeman R, Carlisle A, Chapple T, Curnick D, Dale J, Schallert R, Tickler D, Block B. Shark movement strategies influence poaching risk and can guide enforcement decisions in a large, remote marine protected area. Journal of Applied Ecology. 2020 June 11; 57(9):1782-1792. Available from: https://besjournals.onlinelibrary.wiley.com/doi/10.1111/1365-2664.13654 DOI: 10.1111/1365-2664.13654
3. Carlisle Aaron B, Tickler David, Dale Jonathan J, Ferretti Francesco, Curnick David J, Chapple

Taylor K, Schallert Robert J, Castleton Michael, Block Barbara A. Estimating space use of mobile fishes in a large marine protected area with methodological considerations in acoustic array design. Frontiers in Marine Science. 2019; 6:256.

4. White Timothy D, Carlisle Aaron B, Kroodsma David A, Block Barbara A, Casagrandi Renato, De Leo Giulio A, Gatto Marino, Micheli Fiorenza, McCauley Douglas J. Assessing the effectiveness of a large marine protected area for reef shark conservation. Biological Conservation. 2017; 207:64--71.
5. White TD, Ferretti F, Kroodsma DA, Hazen EL, Carlisle AB, Scales KL, Bograd SJ, Block BA. Predicted hotspots of overlap between highly migratory fishes and industrial fishing fleets in the northeast Pacific. Sci Adv. 2019 Mar;5(3):eaau3761. PubMed Central PMCID: PMC6415957.

Other Significant Products, Whether or Not Related to the Proposed Project

1. Carlisle Aaron B, Kim Sora L, Semmens Brice X, Madigan Daniel J, Jorgensen Salvador J, Perle Christopher R, Anderson Scot D, Chapple Taylor K, Kanive Paul E, Block Barbara A. Using stable isotope analysis to understand the migration and trophic ecology of northeastern Pacific white sharks (Carcharodon carcharias). PloS one. 2012; 7(2):e30492.
2. Carlisle Aaron B, Goldman Kenneth J, Litvin Steven Y, Madigan Daniel J, Bigman Jennifer S, Swithenbank Alan M, Kline Jr Thomas C, Block Barbara A. Stable isotope analysis of vertebrae reveals ontogenetic changes in habitat in an endothermic pelagic shark. Proceedings of the Royal Society B: Biological Sciences. 2015; 282(1799):20141446.
3. Aaron B. Carlisle, Elizabeth Andruszkiewicz Allan, Sora L. Kim, Lauren Meyer, Jesse Port, Stephen Scherrer, John O'Sullivan. Integrating multiple chemical tracers to elucidate the diet and habitat of Cookiecutter Sharks. Scientific Reports. 2021 June; 11(1). Available from: https://doi.org/10.1038%2Fs41598-021-89903-z DOI: 10.1038/s41598-021-89903-z
4. Madigan Daniel J, Carlisle Aaron B, Gardner Luke D, Jayasundara Nishad, Micheli Fiorenza, Schaefer Kurt M, Fuller Daniel W, Block Barbara A. Assessing niche width of endothermic fish from genes to ecosystem. Proceedings of the National Academy of Sciences. 2015; 112(27):8350--8355.
5. Madigan Daniel J, Litvin Steven Y, Popp Brian N, Carlisle Aaron B, Farwell Charles J, Block Barbara A. Tissue turnover rates and isotopic trophic discrimination factors in the endothermic teleost, Pacific bluefin tuna (Thunnus orientalis). PLoS One. 2012; 7(11):e49220.

Certification:

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Certified by Carlisle, Aaron in SciENcv on 2026-10-22 13:12:32

CURRENT AND PENDING (OTHER) SUPPORT INFORMATION

Provide the following information for the Senior/key personnel and other significant contributors.
Follow this format for each person.

*NAME: Bigman, Jennifer

PERSISTENT IDENTIFIER (PID) OF THE SENIOR/KEY PERSON: https://orcid.org/0000-0001-8070-3061

*POSITION TITLE: Assistant Professor/School of Marine Science and Policy

*ORGANIZATION AND LOCATION: University of Delaware, Lewes, Delaware, United States

Proposals/Active Projects

***Proposal/Active Project Title:**	THIS PROPOSAL: Building a dynamic ocean management tool to minimize fisheries bycatch along the U.S. East Coast
***Status of Support:**	Pending
Proposal/Award Number:	27UDRF-SI
***Source of Support:**	University of Delaware Research Foundation
***Primary Place of Performance:**	University of Delaware
***Proposal/Active Project Start Date: (MM/YYYY):**	09/2027
***Proposal/Active Project End Date: (MM/YYYY):**	08/2029
***Total Anticipated Proposal/Project Amount:**	$35,000

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2028	1
2029	1

***Overall Objectives:**

We propose the development of a novel bycatch reduction tool for high-value Mid-Atlantic fisheries that addresses fundamental ecological questions (drivers of geographic distribution) but has societal benefits (maximizing fishing opportunities). Our aims are to (1) identify environmental conditions associated with concentrations of target and bycatch species and (2) quantify how their distributions overlap and vary together across space and time. The resulting predictive maps dynamically identify potential opportunities to reduce bycatch while maintaining access to target species at time scales that align management, species movement and fisheries.

***Statement of Potential Overlap:**

None.

***Proposal/Active Project Title:**	Understanding and predicting ecological dynamics of marine predators through field-based measurements of energy use
***Status of Support:**	Pending
Proposal/Award Number:	27A00194
***Source of Support:**	National Science Foundation
***Primary Place of Performance:**	University of Delaware
***Proposal/Active Project Start Date: (MM/YYYY):**	08/2027
***Proposal/Active Project End Date: (MM/YYYY):**	07/2030
***Total Anticipated Proposal/Project Amount:**	$1,368,065

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2028	0.12
2029	1
2030	1

***Overall Objectives:**

This study has three primary objectives: (1) provide the first, direct measurements of energy use (FMR) in a free-swimming, water-breathing animal; (2) characterize how energy use varies across time (in contrast to traditional measurements which are a snapshot in a controlled lab setting), environments, and behaviors; and (3) assess whether more realistic measures of energy use (FMR) improve predictions of ecological processes (growth and habitat suitability) based on metabolic ecology.

***Statement of Potential Overlap:**

There is no overlap with existing or pending projects.

***Proposal/Active Project Title:**	Inclusive causal modeling to develop climate-resilient fisheries
***Status of Support:**	Pending
Proposal/Award Number:	26A01555
***Source of Support:**	Marine Institute Memorial University/NFRF
***Primary Place of Performance:**	University of Delaware
***Proposal/Active Project Start Date: (MM/YYYY):**	04/2027
***Proposal/Active Project End Date: (MM/YYYY):**	03/2030
***Total Anticipated Proposal/Project Amount:**	$258,088

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2027	1
2028	1
2029	1

***Overall Objectives:**

TBD

***Statement of Potential Overlap:**

TBD

***Proposal/Active Project Title:** Deciphering Environmental Change Impacts and Pathways of Mercury Exposure in Arctic Food Webs: Insights from Marine Predators

***Status of Support:** Pending

Proposal/Award Number: 26A00824

***Source of Support:** National Science Foundation

***Primary Place of Performance:** University of Delaware

***Proposal/Active Project Start Date: (MM/YYYY):** 02/2027

***Proposal/Active Project End Date: (MM/YYYY):** 01/2030

***Total Anticipated Proposal/Project Amount:** $999,977

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2027	1
2028	1
2029	1

***Overall Objectives:**

This project aims to investigate the drivers of temporal changes in an Arctic sentinel species – Eastern Beaufort Sea belugas - using a multifaceted approach combining chemical tracers, existing data, and food-web bioaccumulation modeling. We will generate essential empirical data to identify the past and present dietary mercury sources of this beluga population and construct environmental scenarios in the mercury

bioaccumulation model to explain temporal variability in the beluga MeHg burden.
The results will provide new insights into the environmental factors and pathways driving Arctic ecosystem dynamics and mercury variability, particularly for species that also highly rely on Arctic cod, such as various seabirds, ringed seals, and narwhals.

***Statement of Potential Overlap:**

None

***Proposal/Active Project Title:**	Identifying and improving conservation and monitoring of Atlantic horseshoe crab spawning dynamics in Delaware Bay
***Status of Support:**	Pending
Proposal/Award Number:	26A01129
***Source of Support:**	National Fish and Wildlife Foundation
***Primary Place of Performance:**	University of Delaware
***Proposal/Active Project Start Date: (MM/YYYY):**	01/2027
***Proposal/Active Project End Date: (MM/YYYY):**	12/2029
***Total Anticipated Proposal/Project Amount:**	$498,971

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2027	3.5
2028	3.5
2029	3.5

***Overall Objectives:**

This project will quantify how environmental change and coastal hazards are affecting horseshoe crab spawning dynamics in Delaware Bay and evaluate remote-sensing approaches to improve long-term spawning monitoring. The project will assess trends in spawning timing, abundance, and spatial distribution; identify environmental and physical drivers such as warming, sea-level rise, erosion, flooding, and storm activity; and test whether tools such as fixed cameras, drones, and side-scan sonar can expand survey coverage and improve monitoring resilience. The work will support more effective conservation and management of horseshoe crabs, migratory shorebirds, and Delaware Bay coastal habitats under changing environmental conditions.

***Statement of Potential Overlap:**

TBD

Certification:

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Certified by Bigman, Jennifer in SciENcv on 2026-10-26 08:28:28

CURRENT AND PENDING (OTHER) SUPPORT INFORMATION

Provide the following information for the Senior/key personnel and other significant contributors.
Follow this format for each person.

*NAME: Oliver, Matthew J

PERSISTENT IDENTIFIER (PID) OF THE SENIOR/KEY PERSON: https://orcid.org/0000-0002-6576-914X

*POSITION TITLE: Patricia & Charles Robertson Distinguished Professor of Marine Science & Policy

*ORGANIZATION AND LOCATION: University of Delaware, Lewes, Delaware, United States

Proposals/Active Projects

***Proposal/Active Project Title:** Sustaining and Strengthening the Mid-Atlantic IOOS Enterprise: Advancing Observations to Products and Decision Support

***Status of Support:** Current

Proposal/Award Number: 26A01203

***Source of Support:** National Oceanic and Atmospheric Administration

***Primary Place of Performance:** University of Delaware

***Proposal/Active Project Start Date: (MM/YYYY):** 09/2026

***Proposal/Active Project End Date: (MM/YYYY):** 08/2031

***Total Anticipated Proposal/Project Amount:** $25,000,000

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2027	1.21
2028	1.58
2029	1.49
2030	1.42
2031	1.35

***Overall Objectives:**

The overall objectives are to provide regional satellite products and regional glider products to the MARACOOS.

***Statement of Potential Overlap:**

None.

***Proposal/Active Project Title:**	Integrating Optical and Acoustic Scattering Signals to Advance Remote Sensing
***Status of Support:**	Current
Proposal/Award Number:	25A01002/80NSSC25K0462
***Source of Support:**	National Aeronautics and Space Administration
***Primary Place of Performance:**	University of Delaware
***Proposal/Active Project Start Date: (MM/YYYY):**	09/2025
***Proposal/Active Project End Date: (MM/YYYY):**	08/2028
***Total Anticipated Proposal/Project Amount:**	$150,000

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.01
2026	0.01
2027	0.01

***Overall Objectives:**

This study will: 1) quantify the relationship between optical and acoustic backscatter to assess whether optical techniques reliably detect DVM, 2) evaluate environmental drivers influencing optical-acoustic correlations, 3) compare in-situ glider optical backscatter with HSRL-2 backscatter to validated spaceborne observations of subsurface scattering layers, 4) assess the phytoplankton community composition during episodic carbon export events using PACE satellite products and glider data. This project will enhance the interpretation of optical backscatter signals improving the application of remote sensing techniques for tracking subsurface biological processes.

***Statement of Potential Overlap:**

None.

***Proposal/Active Project Title:**	Advancing oceanographic retrievals for NASA airborne LiDAR systems
***Status of Support:**	Current
Proposal/Award Number:	25A00628/80NSSC25M0045
***Source of Support:**	National Aeronautics and Space Administration
***Primary Place of Performance:**	University of Delaware
***Proposal/Active Project Start Date: (MM/YYYY):**	07/2025
***Proposal/Active Project End Date: (MM/YYYY):**	06/2028

***Total Anticipated Proposal/Project Amount:** $749,785

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	1.8
2027	1.75
2028	0.52

***Overall Objectives:**

The objectives of this proposal is to develop the in-water LiDAR capabilities and interpretation in the Delaware EPSCoR Jurisdiction by providing detailed vertical profiles of phytoplankton, zooplankton and fish that can be matched to NASA HSRL-2 overflights.

***Statement of Potential Overlap:**

There could be some equipment overlap because in both of these proposals, we are requesting funding to purchase a shadowgraph for a slocum glider.

***Proposal/Active Project Title:** Expanding the Geography and User Base of the Atlantic Sturgeon Occurrence Model

***Status of Support:** Current

Proposal/Award Number: 24A01739/80NSSC25K7237

***Source of Support:** National Aeronautics and Space Administration

***Primary Place of Performance:** University of Delaware

***Proposal/Active Project Start Date: (MM/YYYY):** 02/2025

***Proposal/Active Project End Date: (MM/YYYY):** 01/2028

***Total Anticipated Proposal/Project Amount:** $755,784

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	1
2027	1
2028	1

***Overall Objectives:**

The overall objective of this proposal is to expand the user base and geographical scope of an Atlantic Sturgeon forecasting tool for decision making along the east coast of the Us

***Statement of Potential Overlap:**

None

***Proposal/Active Project Title:**	Assessing the Impact of a Satellite Based Atlantic Sturgeon Forecasting Tool
***Status of Support:**	Current
Proposal/Award Number:	24A01546/80NSSC25K7289
***Source of Support:**	National Aeronautics and Space Administration
***Primary Place of Performance:**	University of Delaware
***Proposal/Active Project Start Date: (MM/YYYY):**	12/2024
***Proposal/Active Project End Date: (MM/YYYY):**	11/2025
***Total Anticipated Proposal/Project Amount:**	$100,013

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2025	1

***Overall Objectives:**

The goal of this project is to assess the effectiveness of an existing satellite based Atlantic sturgeon warning tool

***Statement of Potential Overlap:**

None

***Proposal/Active Project Title:**	MARACOOS (Mid-Atlantic IOOS): Powering Understanding and Prediction of the Mid-Atlantic Ocean, Coast, and Estuaries -Glider and Satellite
***Status of Support:**	Current
Proposal/Award Number:	NA21NOS0120096
***Source of Support:**	National Oceanographic and Atmospheric Administration
***Primary Place of Performance:**	University of Delaware
***Proposal/Active Project Start Date: (MM/YYYY):**	08/2021

***Proposal/Active Project End Date: (MM/YYYY):** 07/2026

***Total Anticipated Proposal/Project Amount:** $29,999,997

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2025	1.03
2026	0.12

***Overall Objectives:**

The overall objectives are to provide regional satellite products and regional glider products to the MARACOOS.

***Statement of Potential Overlap:**

None

***Proposal/Active Project Title:** Advancing and integrating animal-borne tags into regional ocean observatories in the Northwest Atlantic

***Status of Support:** Current

Proposal/Award Number: NA20NOS0120221

***Source of Support:** National Oceanographic and Atmospheric Administration

***Primary Place of Performance:** University of Delaware

***Proposal/Active Project Start Date: (MM/YYYY):** 09/2020

***Proposal/Active Project End Date: (MM/YYYY):** 08/2025

***Total Anticipated Proposal/Project Amount:** $1,176,902

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2025	0.12

***Overall Objectives:**

The objective of this project is to develop and deploy CTD tags for sharks in the mid-atlantic.

***Statement of Potential Overlap:**

None

***Proposal/Active Project Title:**	Extreme phytoplankton patches (EPPs) as important carbon export features
***Status of Support:**	Pending
Proposal/Award Number:	26A01745
***Source of Support:**	National Aeronautics and Space Administration
***Primary Place of Performance:**	University of Delaware
***Proposal/Active Project Start Date: (MM/YYYY):**	01/2027
***Proposal/Active Project End Date: (MM/YYYY):**	12/2029
***Total Anticipated Proposal/Project Amount:**	$150,000

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2027	0.01
2028	0.01
2029	0.01

***Overall Objectives:**

The proposed research project seeks to understand the importance of rare extreme phytoplankton patches as drivers of ocean carbon sequestration and biological carbon pump variability.

***Statement of Potential Overlap:**

None.

Certification:

I certify that the information provided is current, accurate, and complete. This includes, but is not limited to, information related to current, pending, and other support (both foreign and domestic) as defined in 42 U.S.C. § 6605.

In accordance with Section 10632 of the CHIPS and Science Act of 2022 (42 U.S.C. § 19232), each individual identified as a senior/key person must certify that they are not a party to a malign foreign talent recruitment program.

Research Security Training Requirement for Federal Award Personnel: In accordance with Section 10634 of the CHIPS and Science Act of 2022 (42 U.S.C. § 19234), each individual identified as a senior/key person must certify that they have completed the requisite research security training that meets the requirements specified in Item 2 of Important Notice No. 149 within 12 months prior to proposal submission.

Certified by Oliver, Matthew in SciENcv on 2026-10-26 11:42:15

CURRENT AND PENDING (OTHER) SUPPORT INFORMATION

Provide the following information for the Senior/key personnel and other significant contributors.
Follow this format for each person.

*NAME: Carlisle, Aaron

PERSISTENT IDENTIFIER (PID): https://orcid.org/0000-0003-0796-6564

*POSITION TITLE: Associate Professor

*ORGANIZATION AND LOCATION: University of Delaware, Newark, Delaware, United States

Proposals and Active Projects

***Proposal/Active Project Title:**	Characterizing the impact of diet on stable isotope dynamics of Dolphinfish
***Status of Support:**	Current
Proposal/Award Number:	26A01215
***Source of Support:**	Monterey Bay Aquarium Research
***Primary Place of Performance:**	University of Delaware, Newark, DE, USA
***Proposal/Active Project Start Date: (MM/YYYY):**	05/2026
***Proposal/Active Project End Date: (MM/YYYY):**	11/2026
***Total Anticipated Proposal/Project Amount:**	$9,000

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.12

***Overall Objectives:**

The objective of this project is to quantify the stable isotope dynamics in captive Dolphinfish at the Monterey Bay Aquarium.

***Statement of Potential Overlap:**

None

***Proposal/Active Project Title:**	Flyway Models and Metrics in the Assessment of OWF Impacts on Migratory Fishes
***Status of Support:**	Current
Proposal/Award Number:	25A01081
***Source of Support:**	UMD Center for Environmental Science/ROSA
***Primary Place of Performance:**	University of Delaware, Newark, DE, USA
***Proposal/Active Project Start Date: (MM/YYYY):**	09/2025
***Proposal/Active Project End Date: (MM/YYYY):**	08/2027
***Total Anticipated Proposal/Project Amount:**	$20,000

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.12

***Overall Objectives:**

The objectives of this project are to: 1) Obtain access permissions and compile databases for sand tiger shark, Atlantic sturgeon, and striped bass telemetry records for the period 2006-2021. 2) Develop and evaluate multi-species and single species regional flyways models. Evaluate flyways for geo-rotational biases and make adjustments. Test for annual and seasonal effects of oceanographic and climate variables. 3) Through feature engineering (machine-learning), extract flyway metrics. 4) Through research partner engagement, evaluate a set of feasible flyway metrics for test species: those that are (1) unobscured by error and bias, (2) meaningful, and (3) readily communicated. Summarize variance and mean tendencies in metrics. Investigate the responsiveness of flyway metrics to oceanographic and climate drivers

***Statement of Potential Overlap:**

None

***Proposal/Active Project Title:**	Mid-Atlantic MBON: Dynamic Biodiversity and Telemetry Data for a Changing Coast
***Status of Support:**	Current
Proposal/Award Number:	-NA24NOSX012C0021-T1-01
***Source of Support:**	National Oceanic Atmospheric Administration
***Primary Place of Performance:**	University of Delaware, Newark, DE, USA
***Proposal/Active Project Start Date: (MM/YYYY):**	07/2024

***Proposal/Active Project End Date: (MM/YYYY):** 06/2028

***Total Anticipated Proposal/Project Amount:** $1,750,000

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.12
2027	0.12
2028	0.12

***Overall Objectives:**

The overall objective of this project is to integrate cutting edge biological (e.g. environmental DNA [eDNA], telemetry, passive acoustics) information with traditional MARACOOS strengths in ocean observing (e.g. high-frequency radar, satellites, models), leading to the development of new data products, analytical tools, and information resources supporting ocean and coastal management in the Mid-Atlantic region.

***Statement of Potential Overlap:**

None

***Proposal/Active Project Title:** Sandbridge Highly Migratory Species: Fish Distribution on a Dredged Shoal

***Status of Support:** Current

Proposal/Award Number: M21AC00008-00

***Source of Support:** Department of Interior

***Primary Place of Performance:** University of Delaware

***Proposal/Active Project Start Date: (MM/YYYY):** 05/2021

***Proposal/Active Project End Date: (MM/YYYY):** 12/2027

***Total Anticipated Proposal/Project Amount:** $888,170

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.5
2027	0.8

***Overall Objectives:**

The goal of this project is to conduct a multidisciplinary field study to characterize the use of Sandbridge Shoal by high priority highly migratory species. This study will provide a robust characterization of how, when, and why HMS use the study site, what their ecological role is in the area, and what life history functions Sandbridge Shoal plays in the life history of these important species.

***Statement of Potential Overlap:**

None

***Proposal/Active Project Title:**	Understanding and predicting ecological dynamics of marine predators through field-based measurements of energy use
***Status of Support:**	Pending
Proposal/Award Number:	27A00194
***Source of Support:**	National Science Foundation
***Primary Place of Performance:**	University of Delaware, Newark, DE, USA
***Proposal/Active Project Start Date: (MM/YYYY):**	08/2027
***Proposal/Active Project End Date: (MM/YYYY):**	07/2030
***Total Anticipated Proposal/Project Amount:**	$1,368,065

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2028	1
2029	1
2030	0.5

***Overall Objectives:**

This study has three primary objectives: (1) provide the first, direct measurements of energy use (FMR) in a free-swimming, water-breathing animal; (2) characterize how energy use varies across time (in contrast to traditional measurements which are a snapshot in a controlled lab setting), environments, and behaviors; and (3) assess whether more realistic measures of energy use (FMR) improve predictions of ecological processes (growth and habitat suitability) based on metabolic ecology

***Statement of Potential Overlap:**

None

***Proposal/Active Project Title:**	Rooted Resilience: Establishing Large Scale Submerged Aquatic Vegetation Restoration Efforts (DE)
***Status of Support:**	Pending
Proposal/Award Number:	25A01409
***Source of Support:**	National Fish & Wildlife Foundation
***Primary Place of Performance:**	University of Delaware
***Proposal/Active Project Start Date: (MM/YYYY):**	03/2026
***Proposal/Active Project End Date: (MM/YYYY):**	02/2030
***Total Anticipated Proposal/Project Amount:**	$1,024,340

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.06
2027	0.06
2028	0.06
2029	0.06

***Overall Objectives:**

This project aims to mitigate key coastal hazard risks including shoreline erosion, storm impacts, and tidal wetland loss due to sea level rise in Delaware's (DE) bays—particularly within shallow, tidally influenced areas. The absence of submerged aquatic vegetation (SAV), including widgeon grass (Ruppia maritima), sago pondweed (Stuckenia pectinata), and eelgrass (Zostera marina), has compromised the region's natural capacity to attenuate wave energy, stabilize sediments, sequester carbon, and protect critical habitats. These lost functions elevate risk to both community infrastructure and vital ecosystems. Restored SAV meadows buffer wave action and erosion, offering cost-effective protection to shoreline infrastructure like Delaware Route 1, residential properties, and water access points vital to Delaware's $4B coastal economy (Rising & Wilson, 2024). These habitats also improve water quality and serve as nursery grounds for
American Eel, Blue Crabs, and Summer Flounder, recreationally and environmentally important species. Once established, SAV meadows will act as seed sources for future restoration efforts.

***Statement of Potential Overlap:**

None

Certification:

I certify that the information provided is current, accurate, and complete. This includes but is not limited to current, pending, and other support (both foreign and domestic) as defined in 42 U.S.C. § 6605.

I also certify that, at the time of submission, I am not a party to a malign foreign talent recruitment program.

In accordance with section 10634 of the CHIPS and Science Act of 2022 (42 U.S.C. § 19234), I certify that I have completed the requisite research security training that meets the requirements specified in NASA's research security training requirements guidance within 12 months prior to proposal submission.

Misrepresentations and/or omissions may be subject to prosecution and liability pursuant to, but not limited to, 18 U.S.C. §§ 287, 1001, 1031 and 31 U.S.C. §§ 3729-3733 and 3802.

Certified by Carlisle, Aaron in SciENcv on 2026-10-23 10:10:24